LAB 5 REPORT

Constrained Dynamics. Holonomic constraints

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1. Introduction

The goal of today's lab is to simulate a double pendulum (again:)), but using constrained dynamics.

Submit this report and code to Moodle.

2. Simulation

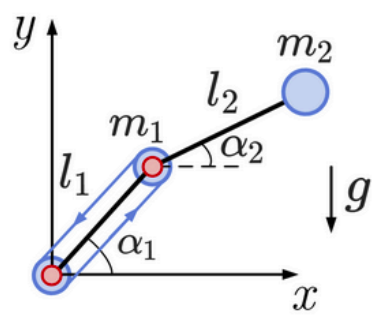
<u>Tasks</u>:

2.1. Derive the dynamics of the double pendulum

- Write the free body motion of each points
- Set the holonomic constraints and obtain the Jacobians
- Plug in all necessary matrices to the equation of motion subject to constraints:

$$\begin{bmatrix} \mathbf{M} & \mathbf{J}^T \\ \mathbf{J} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{Q} - \mathbf{h} \\ -\left(\frac{\partial \dot{\varphi}}{\partial \mathbf{q}}\right)\dot{\mathbf{q}} \end{bmatrix}$$

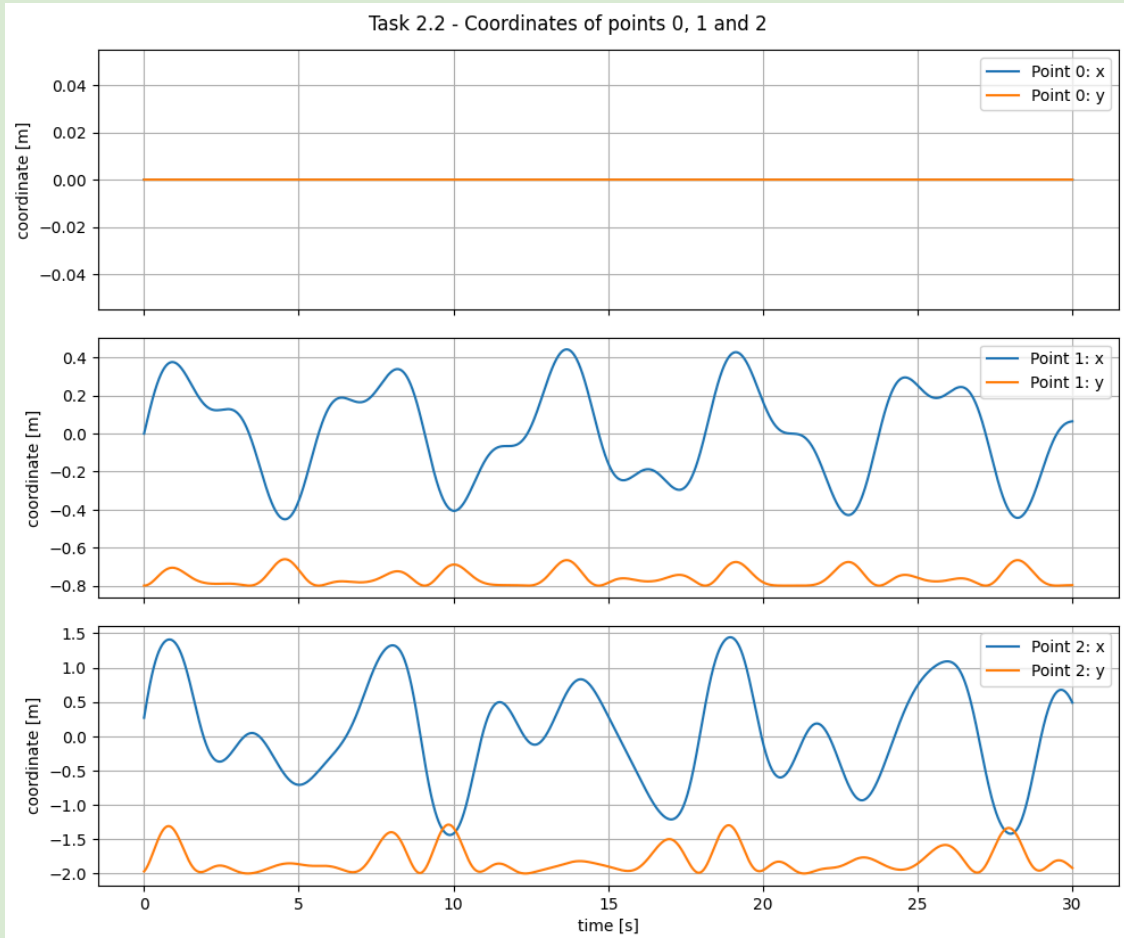
$m_1 = 1$kg
$m_2 = 1$kg
$l_1 = 0.8$ m
$l_2 = 1.2$ m
$g = 9.8$ m/s^2



SIMULATIONS_link

2.2. Simulate the double-link pendulum and visualize the motion (use visualize.py from lab 4). Display in the animation the lengths of each link. Plot the coordinates of the points 0, 1 and 2 versus time.

Note: Add a point 0 to visualize.py, this will help you to solve the next tasks.

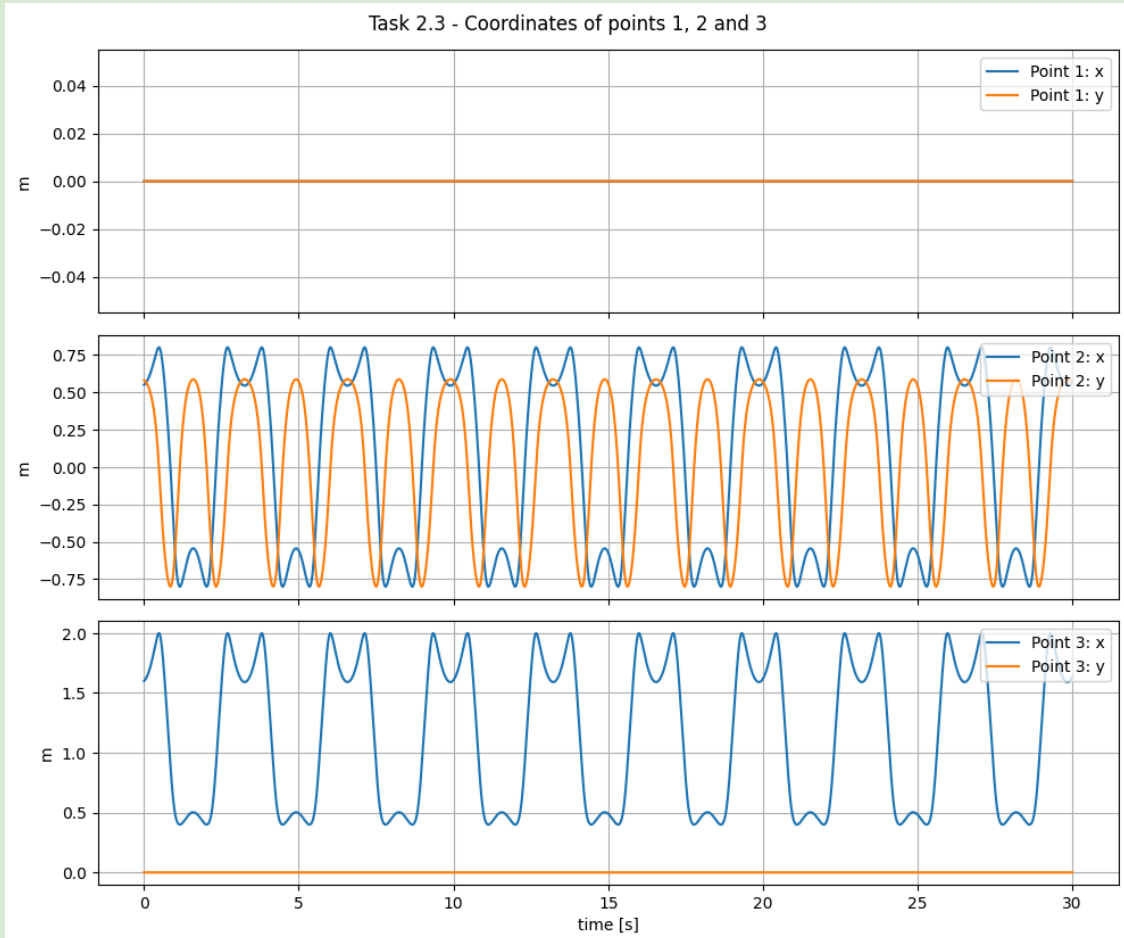


Plots for Task 2.2

Explain the obtained result.

Answer(s): Point zero is fixed pivot. Point/joint 1 and end point oscillate while tip has wider swing than joint 1 because it depends on both angles.

2.3. Implement the dynamics, where the base point should be fixed and the end-effector can move only along the horizontal axis. Plot the coordinates of the points 1, 2 and 3.

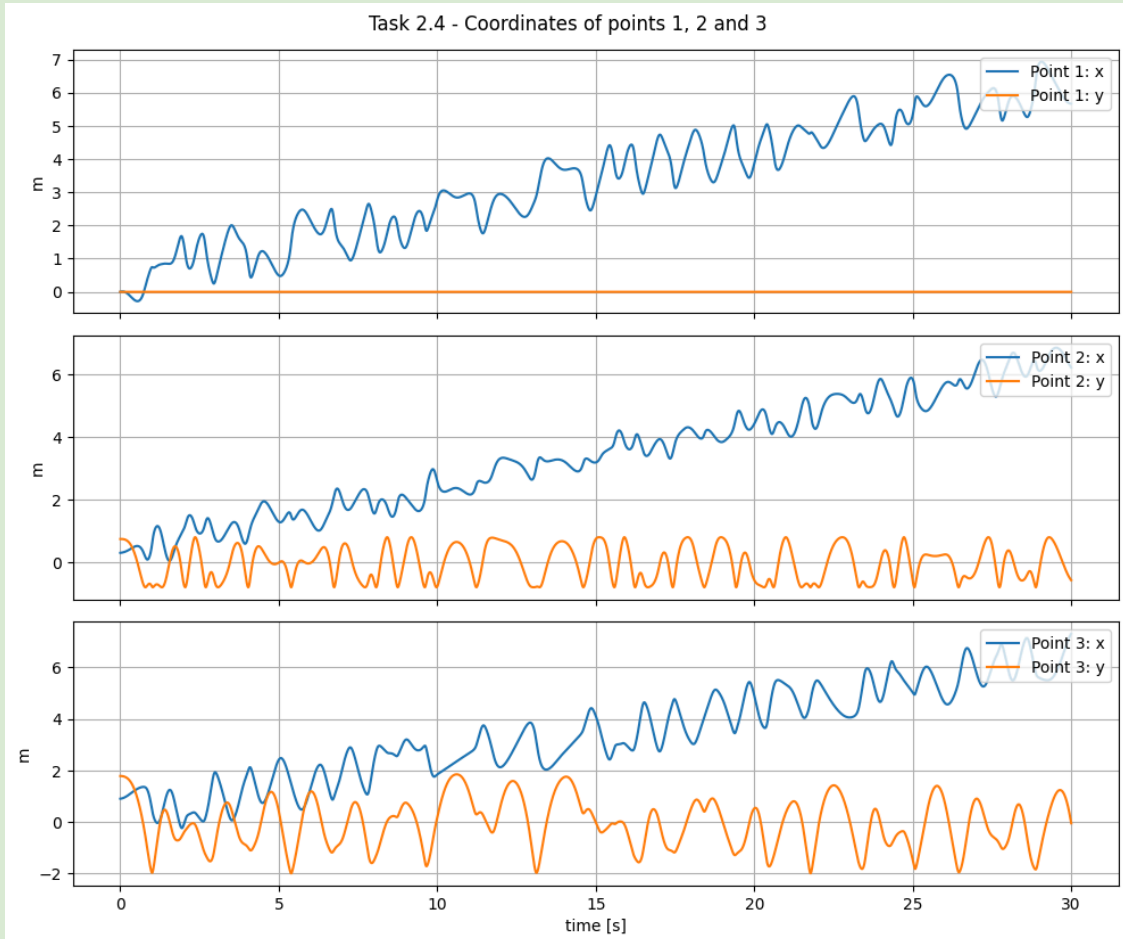


Plots for Task 2.3

How did you set the constraints?

Answer(s):
Two distance constraints |r_2|= l_1, |r_3 - r_2| = l_2 plus (y_3=0); base fixed by omitting (x_1,y_1)) from (q).

2.4. Implement the dynamics, where the base point moves only horizontally. Plot the coordinates of the points 1, 2 and 3.

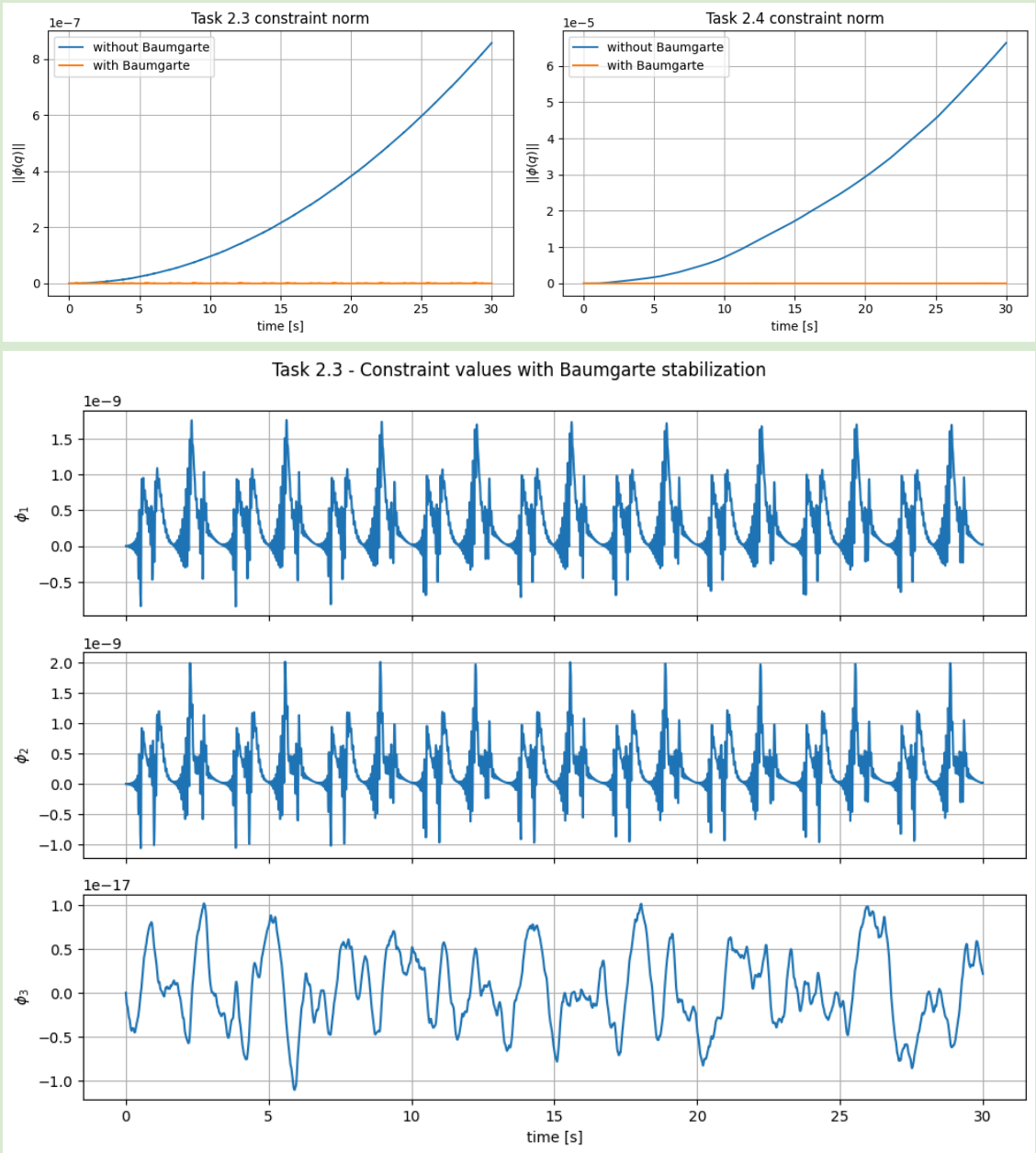


Plots for Task 2.4

How did you set the constraints?

Answer(s):
Impose y1 = 0 for the base line and keep both link-length constraints in half squared-distance form

2.5. Implement Baumgarte stabilization for the tasks 2.3 and 2.4.



Plots for Task 2.5

Explain the obtained result.

Answer(s):

Without Baumgarte stabilization, enforcing constraints only at the acceleration level during numerical integration leads to drift of $\phi(q)$ away from zero. The constraint norm $\|\phi\|$ grows over 30 s (often roughly quadratic in time), reaching about 10^{-7} in Task 2.3 and a larger magnitude (about 10^{-5}) in Task 2.4 because the sliding base adds degrees of freedom and coupling.

Component plots show ϕ_1 and ϕ_2 drifting upward over time, ϕ_3 may stay closer to roundoff in some runs if that row of J is weakly excited.

With Baumgarte parameters $\alpha = \beta = 8$, the acceleration constraint is modified to $J\ddot{q} = -\dot{J}\dot{q} - 2\alpha J\dot{q} - \beta^2\phi$, feeding back position and velocity violations: then $\|\phi\|$ stays near zero on the plots and each ϕ_i remains at numerical noise.